Project Development Phase
Snowflake Project - Data Engineering and SQL Solution

Date	23 October 2026
Team ID	0157CY241012
Project Name	Electric Vehicle (EV) Industry Analytics and Market Intelligence
Maximum Marks	5 Marks

This section evaluates the quality and completeness of the implemented Snowflake solution. The submission demonstrates that the EV market data has been prepared, cleaned, transformed, validated, and made ready for downstream business intelligence analysis.

Instructions Compliance:

1. SQL scripts are modularized into staging (STG), dimensional (DIM), and fact (FACT) layers following Kimball star-schema best practices.
2. Automated data hygiene transformations convert raw heterogeneous telemetry units into standardized metrics (Wh/km, km range, Price/km).
3. Complete DDL, DML, and view definitions are version-controlled in the project repository.
4. Automated data quality constraints guarantee zero nulls in primary business keys.

Solution Summary:

Field	Details
Project / Repository Link	https://github.com/aku797473/data-project1
Data Source(s) Link	https://github.com/aku797473/data-project1/blob/main/data/ev_market_data_cleaned.csv
Database	EV_MARKET_INTELLIGENCE_DB (Schemas: RAW_INGEST, ANALYTICS_CORE, BI_REPORTING)
Warehouse	ANALYTICS_WH (Size: X-Small, Auto-suspend: 60s, Auto-resume: Enabled, Multi-cluster standard)
Tables / Views Created	Tables: RAW_EV_MODELS, DIM_MANUFACTURER, DIM_BATTERY_TECH, DIM_VEHICLE_CLASS, FACT_EV_SALES_METRICS. Views: VW_OEM_BENCHMARKING, VW_BATTERY_CHEMISTRY_CORRELATION, VW_CO2_OFFSET_AGGREGATE.
Key Transformations / Calculations	1. Energy Efficiency normalization: Wh_per_km = (Battery_Capacity_kWh * 1000) / Range_km. 2. Price Affordability Index: Price_per_Km = Selling_Price_USD / Range_km. 3. Environmental Quantification: Annual_CO2_Saved_Kg = (ICE_Baseline_Emissions - EV_Emissions) * Annual_Avg_Mileage.

	4. Window functions for OEM market share rankings.
KPIs created	Average Driving Range (452 km), Average Battery Capacity (68.4 kWh), Mean Energy Efficiency (168 Wh/km), Top OEM Market Share (Tesla 27.4%, BYD 21.2%), Cumulative Tailpipe CO2 Abatement (2.84M Metric Tons).
Key Insights	LFP batteries offer highest consumer ROI for city transit ($74/km range); Solid-State batteries achieve break-through driving autonomy (>750 km); Electric SUVs generate 42% of total industry revenue.

Snowflake Quality Checklist:

S.No	Criteria	Status (Yes / No)
1	Required data is successfully loaded and accessible in Snowflake	Yes
2	Data is cleaned, transformed, and prepared appropriately	Yes
3	Required tables, views, and data relationships are correctly implemented	Yes
4	Required SQL logic, calculations, and KPIs are correctly implemented	Yes
5	Data and solution outputs are validated against project requirements	Yes
6	Solution is properly organized, documented, and functional	Yes

Additional Notes / Comments:
All SQL data pipeline definitions and warehouse configuration queries have been validated for idempotency, ANSI SQL compliance, sub-second query latency, and zero data leakage. Data extracts are mirrored into Tableau Public.